

Chapter 6

Preference, utility, and choice when alternatives are uncertain prospects

We consider agents in a world in which alternatives are “uncertain prospects.” An example is an agent who chooses between two lottery tickets that give him/her different payments with different odds. Another example is an agent who decides, when leaving home in the morning, whether to carry an umbrella or not. Carrying the umbrella gives the agent an uncertain level of welfare. If it rains, the umbrella will come handy and the benefit of not getting wet will compensate the effort of carrying it. If it does not rain, the umbrella will just occupy space in his/her bag without any benefit.

In contrast to the theory developed in the previous lectures, here it is not trivial what the set of alternatives is. In abstract choice theory alternatives are elements of a set X . In consumer theory, alternatives are bundles of goods, i.e., elements of some Euclidean space. What mathematical objects represent uncertain prospects? One has to solve this question before talking about choice and preference over this sort of alternatives.

Let us remark that since abstract choice theory is general, then all results in Chapter 1 apply to choice and preference when alternatives are uncertain prospects. Parallel to consumer theory, our objective here is to impose more structure in the model, so we can better characterize the behavior of agents in these uncertain situations.

6.1 The mathematical structure of an uncertain prospect

The first uncertain prospects that we study are the “risky prospects.” We assume that there is a set of possible outcomes that we denote by Z . This set is different from the set of alternatives. A **risky prospect**, or an alternative, is a lottery over the outcomes Z , i.e., a “probability measure” or a “distribution” on Z . We denote the set of risky prospects over Z by $\Delta(Z)$.

Risky prospects are a natural and handy representation for many uncertain prospects. These are all the prospects for which it is reasonable that the agent envisions some outcomes happening with some probabilities. For the moment, we will not inquire where these probabilities come from. We will just assume that they exist and that the agent accepts them as true.

The preceding two paragraphs seem to leave open the possibility that some uncertain prospects do not fit into the mathematical structure of a risky prospect. It is not trivial to think about one that does not. Nevertheless, not only are there some, but also they seem to play an important role in agents behavior. Big part of the modern literature on decision theory has been motivated by the existence of these uncertain prospects that cannot be represented by risky prospects. We will come back to this issue later.

Let us return to the study of risky prospects. For the sake of concreteness and mathematical simplicity we study them when the set of outcomes is either a finite set or the Real line. In these two cases, the members of $\Delta(Z)$ are familiar to us.

6.1.1 Finite set of outcomes

Suppose that $Z \equiv \{z_1, \dots, z_n\}$ for some $n \in \mathbb{N}$. A probability measure or distribution on Z is a function $\pi : Z \rightarrow [0, 1]$ such that for each $z \in Z$, $\pi(z) \geq 0$ and $\sum_{z \in Z} \pi(z) = 1$. The set of probability measures on Z is $\Delta(Z)$. In order to simplify notation, for each $i \in \mathbb{N}$ we write we write π_i instead of $\pi(z_i)$.

A risky prospect over a finite set of outcomes can be graphically represented by a rooted tree. Final nodes are assigned to outcomes. The probability of each outcome is the probability with which that final node in the tree is reached.

If $Z \equiv \{z_1, z_2\}$, one can graphically represent $\Delta(Z)$ by means of the segment $[0, 1]$. Each point $x \in [0, 1]$ represents the lottery $\pi_1 = x$ and $\pi_2 = 1 - x$. Conversely, each lottery π is represented by the point $\pi_1 \in [0, 1]$.

Analogously, if $Z \equiv \{z_1, z_2, z_3\}$, one can graphically represent $\Delta(Z)$ by an equilateral triangle with height 1 and whose vertices are labeled z_1 , z_2 , and z_3 . A point x in the triangle represents the lottery for which π_1 is the distance between x and the side that opposes vertex z_1 , π_2 is the distance between x and the side that opposes vertex z_2 , and π_3 is the distance between x and the side that opposes vertex z_3 (the summation of the distances to the sides from each point inside an equilateral triangle is a constant). Conversely, each lottery π is represented by the point whose distances to the respective sides are π_1 , π_2 and π_3 (one should be able to prove that given an arbitrary π , there is such a point).

Definition 112. Let $u : Z \rightarrow \mathbb{R}$ and $\pi \in \Delta(Z)$. The **expected value of u with respect to π** is

$$E_\pi u \equiv \sum_{z \in Z} u(z) \pi(z).$$

Definition 113. Let $\{\alpha_i\}_{i=1}^K$ be a list of non-negative numbers such that $\sum_{i=1}^K \alpha_i = 1$. Let $\{\pi^i\}_{i=1}^K$ be a list of elements of $\Delta(Z)$. The **compound lottery of $\{\pi^i\}_{i=1}^K$ with weights $\{\alpha_i\}_{i=1}^K$** is the element of $\Delta(Z)$, denoted by $\alpha_1 \pi^1 + \dots + \alpha_K \pi^K$, defined as follows. For each $z \in Z$,

$$\alpha_1 \pi^1 + \dots + \alpha_K \pi^K(z) \equiv \sum_{i=1}^K \alpha_i \pi^i(z).$$

6.1.2 Money lotteries

We also consider the case in which outcomes are Real numbers, i.e., $Z \equiv \mathbb{R}$. We interpret these outcomes as monetary values. We refer to the probability measures on Z as money lotteries. Formally, an element in $\Delta(Z)$ is a probability measure on the measure space $(\mathbb{R}, \mathcal{F})$ where $\mathcal{F}$ is an appropriate sigma-algebra. In practice, we can make do by representing money lotteries by their cumulative distribution function (c.d.f), i.e., the function that assigns to each $x \in \mathbb{R}$, the probability that the outcome turns out to be less than or equal to x , i.e., $\text{Prob}(z \leq x)$. The generic c.d.f is denoted by $F \in \Delta(Z)$. A c.d.f F admits a density function, denoted by $f : \mathbb{R} \rightarrow \mathbb{R}$, if for each $x \in \mathbb{R}$,

$$F(x) = \int_{-\infty}^x f(z) dz.$$

Definition 114. Let $u : \mathbb{R} \rightarrow \mathbb{R}$ and $F \in \Delta(Z)$. The **expected value of u with respect to F** is

$$E_F u \equiv \int_{-\infty}^{+\infty} u dF.$$

Whenever F admits a density function f , then

$$E_F u \equiv \int_{-\infty}^{+\infty} u(z) f(z) dz.$$

Definition 115. The **expected value of F** is

$$\mu_F \equiv \int_{-\infty}^{+\infty} z dF.$$

Whenever F admits a density function f , then

$$\mu_F \equiv \int_{-\infty}^{+\infty} z f(z) dz.$$

Definition 116. Let $\{\alpha_i\}_{i=1}^K$ be a list of non-negative numbers such that $\sum_{i=1}^K \alpha_i = 1$. Let $\{F^i\}_{i=1}^K$ be a list of elements of $\Delta(Z)$. The **compound lottery of $\{F^i\}_{i=1}^K$ with weights $\{\alpha_i\}_{i=1}^K$** is the element of $\Delta(Z)$, denoted by $\alpha_1 F^1 + \cdots + \alpha_K F^K$, defined as follows. For each $z \in Z$,

$$\alpha_1 F^1 + \cdots + \alpha_K F^K(z) \equiv \sum_{i=1}^K \alpha_i F^i(z).$$

6.2 Expected Utility Theory

In this section we follow the axiomatic method in the study of agents behavior. This method consists of two steps. The first is to state a “short” list of axioms, or “simple” principles, that one could expect are satisfied by agents’ behavior. The second is to characterize the family of preferences that satisfy the axioms.

In the remainder of this section Z is a finite set (all result generalize to infinite Z with an appropriate extension of the continuity axiom).

Expected Utility Theory, due to John von Neumann and Oscar Morgenstern, is an axiomatic theory that characterizes the preferences on $\Delta(Z)$ that satisfy the following three axioms.

Let $\succsim$ be a binary relation on $\Delta(Z)$.

Axiom A1 (order): $\succsim$ is complete and transitive.

Axiom A2 (continuity): for each triple $\{\pi, \pi', \pi''\} \subseteq \Delta(Z)$ the sets

$$\{\alpha \in [0, 1] : \alpha\pi + (1 - \alpha)\pi' \succsim \pi''\},$$

and

$$\{\alpha \in [0, 1] : \pi'' \succsim \alpha\pi + (1 - \alpha)\pi'\},$$

are closed.

Axiom A3 (independence): for each triple $\{\pi, \pi', \pi''\} \subseteq \Delta(Z)$ and each $\alpha \in (0, 1)$,

$$\pi \succsim \pi' \text{ if and only if } \alpha\pi + (1 - \alpha)\pi'' \succsim \alpha\pi' + (1 - \alpha)\pi''.$$

How sensible are these axioms?

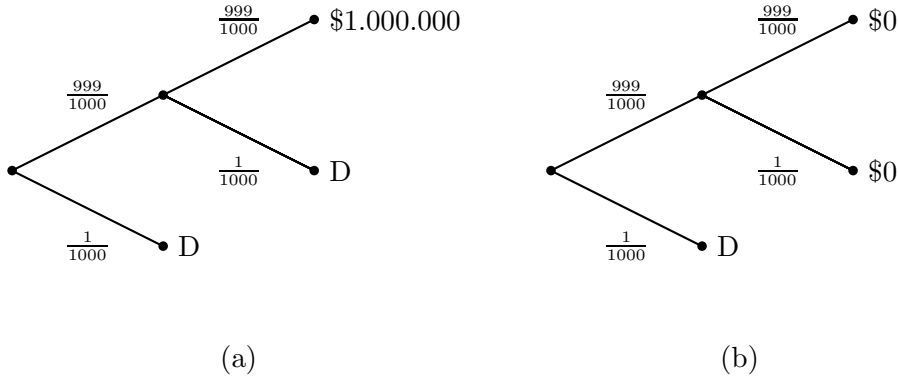


Figure 6.1: Independence axiom

Lemma 4. Let $\succsim$ be a binary relation on $\Delta(Z)$ that satisfies A1 and A3. Then, for each $\{\pi, \pi', \pi''\} \subseteq \Delta(Z)$ and each $\alpha \in (0, 1)$,

$$\pi \succ \pi' \text{ if and only if } \alpha\pi + (1 - \alpha)\pi'' \succ \alpha\pi' + (1 - \alpha)\pi'',$$

and

$$\pi \sim \pi' \text{ if and only if } \alpha\pi + (1 - \alpha)\pi'' \sim \alpha\pi' + (1 - \alpha)\pi''.$$

Proof. Problem 1, Section 6.2.1. □

Lemma 5. Let $\succsim$ be a binary relation on $\Delta(Z)$ that satisfies A1 and A3. Then, for each $\{\pi, \pi', \pi'', \pi'''\} \subseteq \Delta(Z)$ such that $\pi \succ \pi'$ and $\pi'' \succ \pi'''$, and each $\alpha \in [0, 1]$, we have that $\alpha\pi + (1 - \alpha)\pi'' \succ \alpha\pi' + (1 - \alpha)\pi'''$.

Proof. Problem 1, Section 6.2.1. $\square$

Proposition 58. Assume that Z is a finite set. Let $\succsim$ be a binary relation on $\Delta(Z)$ satisfying A1 and A3. Then, there are best and worst lotteries, i.e., $\{\bar{\pi}, \underline{\pi}\} \subseteq \Delta(Z)$ such that for each $\pi \in \Delta(Z)$,

$$\bar{\pi} \succsim \pi \succsim \underline{\pi}.$$

Proof. Problem 6, Section 6.2.1. $\square$

Theorem 21. A binary relation $\succsim$ on $\Delta(Z)$ satisfies A1, A2, A3 if and only if there is an index $u : Z \rightarrow \mathbb{R}$ such that $\succsim$ is represented by the function $U : \Delta(Z) \rightarrow \mathbb{R}$ defined by: for each $\pi \in \Delta(Z)$, $U(\pi) \equiv E_{\pi}u$.

Moreover, u is unique up to positive affine linear transformation, i.e., if there is another function $v : Z \rightarrow \mathbb{R}$ such that $\succsim$ is represented by the function $E_{(\cdot, \pi)}v$, then there are Real numbers a and b such that $a > 0$ and for each $z \in Z$, $u(z) = av(z) + b$.

Proof. By proposition 58, there are lotteries $\{\bar{\pi}, \underline{\pi}\} \subseteq \Delta(Z)$ such that for each $\pi \in \Delta(Z)$,

$$\bar{\pi} \succsim \pi \succsim \underline{\pi}.$$

We can suppose without loss of generality that $\bar{\pi} \succ \underline{\pi}$, for otherwise the theorem is trivial. We will proceed in four steps.

1. Given $\pi \in \Delta(Z)$, we will find a unique $\alpha \in [0, 1]$ such that

$$\pi \sim \alpha\bar{\pi} + (1 - \alpha)\underline{\pi}.$$

2. The function $U : \Delta(Z) \rightarrow \mathbb{R}$ defined by:

$$\begin{aligned} U : \Delta(Z) &\rightarrow \mathbb{R} \\ \pi &\mapsto U(\pi) \equiv \alpha \quad \text{where } \pi \sim \alpha\bar{\pi} + (1 - \alpha)\underline{\pi}, \end{aligned}$$

represents $\succsim$.

3. For each pair of lotteries $\{\pi, \pi'\} \subseteq \Delta(Z)$ and each $\alpha \in [0, 1]$,

$$U(\alpha\pi + (1 - \alpha)\pi') = \alpha U(\pi) + (1 - \alpha)U(\pi').$$

4. Let $u : Z \rightarrow \mathbb{R}$ be the index defined by: for each $z \in Z$, $u(z) \equiv U(\delta_z)$ where δ_z is the lottery that assigns probability one to outcome z . Then, for each $\pi \in \Delta(Z)$,

$$U(\pi) = \sum_{z \in Z} u(z)\pi(z).$$

Step 1: Let $\pi \in \Delta(Z)$. We first prove that there is $\alpha \in [0, 1]$ such that $\pi \sim \alpha\bar{\pi} + (1 - \alpha)\underline{\pi}$. Let

$$A \equiv \{\alpha \in [0, 1] : \pi \succsim \alpha\bar{\pi} + (1 - \alpha)\underline{\pi}\},$$

and

$$B \equiv \{\alpha \in [0, 1] : \alpha\bar{\pi} + (1 - \alpha)\underline{\pi} \succsim \pi\}.$$

Since $0 \in A$ and $1 \in B$, then both sets are non-empty. Since $\succsim$ satisfies A2, then both sets are closed. Since $\succsim$ satisfies A1, then $[0, 1] \subseteq A \cup B$. Since $[0, 1]$ is connected, then $A \cap B \neq \emptyset$. Thus there is $\alpha \in A \cap B$. Thus, $\pi \sim \alpha\bar{\pi} + (1 - \alpha)\underline{\pi}$.

We now prove that if $\pi \sim \alpha\bar{\pi} + (1 - \alpha)\underline{\pi}$ and $\pi \sim \beta\bar{\pi} + (1 - \beta)\underline{\pi}$, then $\alpha = \beta$. This is implied by the following two claims.

Claim 1: if $\bar{\pi} \succ \underline{\pi}$ and $\alpha \in (0, 1)$, then $\bar{\pi} \succ \alpha\bar{\pi} + (1 - \alpha)\underline{\pi} \succ \underline{\pi}$. By Lemma 4, $\alpha\bar{\pi} + (1 - \alpha)\bar{\pi} \succ \alpha\bar{\pi} + (1 - \alpha)\underline{\pi}$ and $\alpha\bar{\pi} + (1 - \alpha)\underline{\pi} \succ \alpha\underline{\pi} + (1 - \alpha)\underline{\pi}$.

Claim 2: if $\alpha > \beta$, then

$$\pi \sim \alpha\bar{\pi} + (1 - \alpha)\underline{\pi} \succ \beta\bar{\pi} + (1 - \beta)\underline{\pi} \sim \pi. \quad (6.1)$$

By Claim 1, $\bar{\pi} \succ \beta\bar{\pi} + (1 - \beta)\underline{\pi}$. Now,

$$\alpha\bar{\pi} + (1 - \alpha)\underline{\pi} = \gamma\bar{\pi} + (1 - \gamma)[\beta\bar{\pi} + (1 - \beta)\underline{\pi}],$$

where $\gamma = \frac{\alpha - \beta}{1 - \beta} \in (0, 1)$. By Claim 1, (6.1) follows. $\triangle$

Step 2: let $U : \Delta(Z) \rightarrow \mathbb{R}$ be defined by:

$$\begin{aligned} U : \Delta(Z) &\rightarrow \mathbb{R} \\ \pi &\mapsto U(\pi) \equiv \alpha \quad \text{where } \pi \sim \alpha\bar{\pi} + (1 - \alpha)\underline{\pi}. \end{aligned}$$

We prove that U represents $\succsim$, i.e., for each pair $\{\pi, \pi'\} \subseteq \Delta(Z)$, $\pi \succsim \pi'$ if and only if $U(\pi) \geq U(\pi')$.

$\Rightarrow$ Assume $\pi \succsim \pi'$. We claim that $U(\pi) \geq U(\pi')$. Suppose by contradiction that $U(\pi') > U(\pi)$. By Claim 2,

$$\pi' \sim U(\pi')\bar{\pi} + (1 - U(\pi'))\underline{\pi} \succ U(\pi)\bar{\pi} + (1 - U(\pi))\underline{\pi} \sim \pi.$$

Since $\succsim$ satisfies A1, then $\pi' \succ \pi$. This is a contradiction.

$\Leftarrow$ Assume $U(\pi) \geq U(\pi')$. If $U(\pi) = U(\pi')$, then

$$\pi \sim U(\pi)\bar{\pi} + (1 - U(\pi))\underline{\pi} \sim U(\pi')\bar{\pi} + (1 - U(\pi'))\underline{\pi} \sim \pi'.$$

If $U(\pi) > U(\pi')$, by Claim 2,

$$\pi \sim U(\pi)\bar{\pi} + (1 - U(\pi))\underline{\pi} \succ U(\pi')\bar{\pi} + (1 - U(\pi'))\underline{\pi} \sim \pi'.$$

Since $\succsim$ satisfies A1, then $\pi \succ \pi'$.

Step 3: for each pair of lotteries $\{\pi, \pi'\} \subseteq \Delta(Z)$ and each $\alpha \in [0, 1]$,

$$U(\alpha\pi + (1 - \alpha)\pi') = \alpha U(\pi) + (1 - \alpha)U(\pi').$$

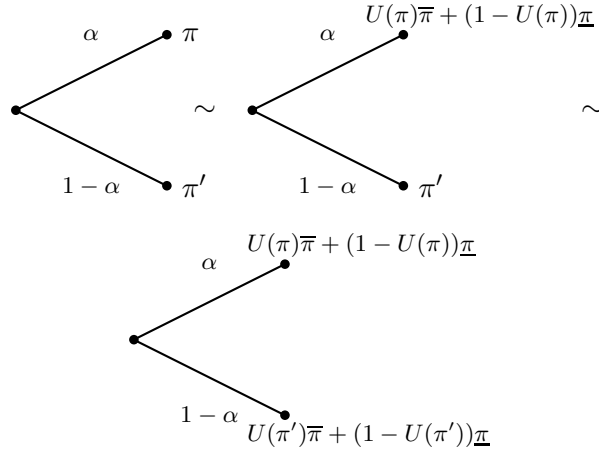


Figure 6.2: Linearity of Expected Utility Representation.

By Lemma 4,

$$\begin{aligned} \alpha\pi + (1 - \alpha)\pi' &\sim \alpha[U(\pi)\bar{\pi} + (1 - U(\pi))\underline{\pi}] + (1 - \alpha)\pi' \\ &\sim \alpha[U(\pi)\bar{\pi} + (1 - U(\pi))\underline{\pi}] + (1 - \alpha)[U(\pi')\bar{\pi} + (1 - U(\pi'))\underline{\pi}]. \end{aligned}$$

Thus,

$$\alpha\pi + (1 - \alpha)\pi' \sim (\alpha U(\pi) + (1 - \alpha)U(\pi'))\bar{\pi} + (1 - (\alpha U(\pi) + (1 - \alpha)U(\pi')))\underline{\pi}.$$

Hence, $U(\alpha\pi + (1 - \alpha)\pi') = \alpha U(\pi) + (1 - \alpha)U(\pi')$.

Step 4: let $u : Z \rightarrow \mathbb{R}$ be the index defined by: for each $z \in Z$, $u(z) \equiv U(\delta_z)$ where δ_z is the lottery that assigns probability one to outcome z . Then, for each $\pi \in \Delta(Z)$,

$$U(\pi) = \sum_{z \in Z} u(z)\pi(z).$$

For the proof see Problem 7, Section 6.2.1.

Step 5: u is unique up to affine linear transformations. For the proof see Problem 8, Section 6.2.1. $\square$

6.2.1 Problems

1. Prove Lemmas 4 and 5.
2. Show that the summation of the distances to the sides from each point in an equilateral triangle is a constant.
3. Let $Z \equiv \{z_1, z_2, z_3\}$. Consider the preference represented by the Expected Utility (EU) associated with index u defined as follows: $u(z_1) = 1$, $u(z_2) = 3$ and $u(z_3) = 100$. Does the EU associated with $100u$ represent the same preference. Does the EU associated with u^2 represent the same preference. For each preference, draw indifference sets in the triangle that represents $\Delta(Z)$ and illustrate your conclusion.
4. Prove that if $\succsim$ is an Expected Utility Maximizer, then satisfies A1 and A3 (state and prove that each axiom is satisfied). Provide an intuition why A2 is also satisfied.
5. Let $Z \equiv \{a, b, c\}$.
 - (a) Construct a preference on $\Delta(Z)$ satisfying A1 and A2 that does not satisfy A3.
 - (b) Construct a preference on $\Delta(Z)$ satisfying A1 and A3 that does not satisfy A2.
 - (c) Construct a preference on $\Delta(Z)$ satisfying A2 and A3 that does not satisfy A1 (provide only the intuition why A2 is satisfied).
6. Let Z be a finite set of prizes and $\succsim$ a preference on $\Delta(Z)$, i.e., the set of lotteries on Z . The generic lottery is denoted $\pi \in \Delta(Z)$.
 - (a) Formally define the Independence axiom (IA), completeness, and transitivity.
 - (b) Prove that if $\succsim$ is complete, transitive, and satisfies IA, then there are best and worst lotteries, i.e., there are $\bar{\pi}, \underline{\pi}$ such that for each $\pi \in \Delta(Z)$,

$$\bar{\pi} \succsim \pi \succsim \underline{\pi}.$$

Do it following these steps: for each $z \in Z$, let δ_z be the probability measure on Z that places probability one in outcome z .

- i. Let $\{z, z'\} \subseteq Z$ be such that $\delta_z \succsim \delta_{z'}$. Prove that for each $\alpha \in [0, 1]$, $\delta_z \succsim \alpha\delta_z + (1 - \alpha)\delta_{z'} \succsim \delta_{z'}$.
 - ii. Prove that it is always possible to enumerate the elements of Z , say $\{z_1, \dots, z_n\}$, in such way that $\delta_{z_1} \succsim \dots \succsim \delta_{z_n}$. Prove, by induction over n , that for each $(\alpha_i)_{i=1}^n \in \mathbb{R}_+^n$ such that $\sum_{i=1}^n \alpha_i = 1$, $\delta_{z_1} \succsim \alpha_1\delta_{z_1} + \dots + \alpha_n\delta_{z_n} \succsim \delta_{z_n}$.
 - iii. Conclude that there are best and worst lotteries, i.e., $\bar{\pi}$ and $\underline{\pi}$ such that for each $\pi \in \Delta(Z)$, $\bar{\pi} \succsim \pi \succsim \underline{\pi}$.
7. Prove Step 4 in the proof of Theorem 21. Hint: use an induction argument over the cardinality of the set $\{i : \pi(z_i) > 0\}$.
8. Prove Step 5 in the proof of Theorem 21. That is, u constructed in the proof is unique up to positive affine linear transformation, i.e., if there is another function $v : Z \rightarrow \mathbb{R}$ such that $\succsim$ is represented by the function $E_{(\pi)}v$, then there are Real numbers a and b such that $a > 0$ and for each $z \in Z$, $u(z) = av(z) + b$.

6.3 Subjective Expected Utility Theory

Primitives of the model:

1. Set of prices Z .
2. State space Ω .
3. Set of all simple acts $\mathcal{F}$, i.e., functions

$$\begin{array}{rcl} f : & \Omega & \rightarrow Z \\ & \omega & \mapsto f(\omega) \end{array}$$

4. Binary relation on $\mathcal{F}$.

Subjective Expected Utility Theory answers the fundamental question: when does an agent have a probability distribution over states and makes decisions using Expected Utility with respect to that probability measure?

The answer is axiomatic.

Theorem 22 (Savage). A binary relation on $\mathcal{F}$ satisfies P1-P6 if and only if there is a convex-ranged probability measure p on Ω and a function $u : Z \rightarrow \mathbb{R}$ non-constant, such that the binary relation is represented by the function that associates with each $f \in \mathcal{F}$ the value

$$U(f) \equiv \int_{\Omega} u(f(\omega)) dp(\omega).$$

Moreover, if (q, v) is another such representing pair, then $p = q$ and v is a positive affine linear transformation of u .

Do people always act as if they assigned probabilities to uncertain outcomes? The example illustrated in Figure 6.3, known as Ellberg Paradox, suggest the opposite.

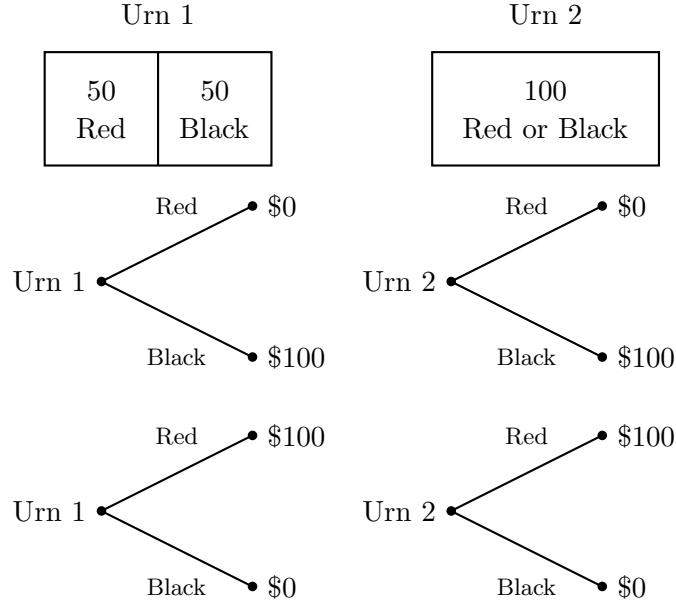


Figure 6.3: Ellsberg Paradox.

6.4 Money lotteries and risk aversion

6.4.1 Expected utility theory

Primitives of the model:

1. Set of prizes $Z \equiv \mathbb{R}$.
2. Alternative set $\Delta(Z)$.

Definition 117. Let $\succsim$ be a preference on $\Delta(\mathbb{R})$. Then, $\succsim$ is **money monotone** if for each $\{z, z'\} \subseteq \mathbb{R}$ such that $z \geq z'$, $\delta_z \succsim \delta_{z'}$; it is **strictly money monotone** if for each $\{z, z'\} \subseteq \mathbb{R}$ such that $z > z'$, $\delta_z \succ \delta_{z'}$.

Theorem 23. A preference on $\succsim$ on $\Delta(\mathbb{R})$ satisfies A1, A2*, and A3* if and only if there is an index $u : \mathbb{R} \rightarrow \mathbb{R}$ such that for each $\{F, G\} \subseteq \Delta(\mathbb{R})$, $F \succsim G$ if and only if

$$E_F u \geq E_G u.$$

The index u is unique up to positive affine linear transformations. Moreover, if $\succsim$ is money monotone, then u is non-decreasing; and if $\succsim$ is strictly money monotone, then u is increasing.

6.4.2 Risk aversion

Consider the lottery F that assigns probability $\frac{1}{2}$ to the outcomes \$1 and -1 . Consider an agent who chooses between F and the lottery that places probability one in \$0, i.e., δ_0 . What would an agent who is “averse to risk” choose? Lottery F offers only risk compared to δ_0 . Thus a risk averse agent would prefer δ_0 to F .

Definition 118. Let $\succsim$ be a preference on $\Delta(\mathbb{R})$. Then, $\succsim$ is:

1. **risk averse** if for each $F \in \Delta(\mathbb{R})$,

$$\delta_{\mu_F} \succsim F.$$

2. **strictly risk averse** if for each $F \in \Delta(\mathbb{R})$ such that $\delta_{\mu_F} \neq F$,

$$\delta_{\mu_F} \succ F.$$

3. **risk neutral** if for each $F \in \Delta(\mathbb{R})$,

$$\delta_{\mu_F} \sim F.$$

6.4.3 Index characterization of risk aversion

Proposition 59. Let $\succsim$ be an EU maximizer with associated index u . Then $\succsim$ is risk averse if and only if u is concave.

Proof. ($\Rightarrow$) Suppose $\succsim$ is risk averse. We prove that u is concave. Let $\{x, y\} \subseteq \mathbb{R}$ and $\alpha \in (0, 1)$. Let F be the lottery that assigns probability α to x and $(1 - \alpha)$ to y . Since $\succsim$ is risk averse then,

$$\delta_{\alpha x + (1-\alpha)y} \succsim F.$$

Thus,

$$U(\delta_{\alpha x + (1-\alpha)y}) = u(\alpha x + (1-\alpha)y) \geq \alpha u(x) + (1-\alpha)u(y) = U(F).$$

Which proves that u is concave.

($\Leftarrow$) Suppose that u is concave. We prove that $\succsim$ is risk averse. Let $F \in \Delta(\mathbb{R})$. We claim that $\delta_{\mu_F} \succsim F$. Consider the tangent to u at μ_F . The equation of this line is for each $t \in \mathbb{R}$,

$$y(t) = u(\mu_F) + u'(\mu_F)(t - \mu_F).$$

Since u is concave, then for each $t \in \mathbb{R}$, $y(t) \geq u(t)$. Thus,

$$\int y(t) dF(t) \geq \int u(t) dF(t).$$

Which is

$$\int (u(\mu_F) + u'(\mu_F)(t - \mu_F)) dF(t) \geq \int u(t) dF(t).$$

Since $\int dF = 1$ and $\int t dF(t) = \mu_F$, then (the following equation is known as Jensen's inequality)

$$u(\mu_F) \geq \int u(t) dF(t). \quad (6.2)$$

Thus, $\delta_{\mu_F} \succsim F$. □

6.4.4 Certainty equivalent and probability premium characterization of risk aversion

Definition 119. Let $\succsim$ be an EU maximizer with associated index u and $F \in \Delta(\mathbb{R})$.

1. The **certainty equivalent of F with respect to u** is the amount of money $c(F, u)$ such that

$$\delta_{c(F, u)} \sim F.$$

Equivalently,

$$u(c(F, u)) = \int u dF.$$

2. Let $x \in \mathbb{R}$ and $\varepsilon > 0$. The **probability premium of at x with respect to ε and u** is the premium $\pi(x, \varepsilon, u)$ such that if F is the lottery that assigns probability $\frac{1}{2} - \pi(x, \varepsilon, u)$ to $x - \varepsilon$ and probability $\frac{1}{2} + \pi(x, \varepsilon, u)$ to $x + \varepsilon$,

$$\delta_x \sim F.$$

Equivalently,

$$u(x) = \left(\frac{1}{2} - \pi(x, \varepsilon, u) \right) u(x - \varepsilon) + \left(\frac{1}{2} + \pi(x, \varepsilon, u) \right) u(x + \varepsilon).$$

Proposition 60. Let $\succsim$ be an EU maximizer with associated index u . Then, the following conditions are equivalent:

1. $\succsim$ is risk averse.
2. For each $F \in \Delta(\mathbb{R})$, $c(F, u) \leq \mu_F$.
3. For each $x \in \mathbb{R}$ and each $\varepsilon > 0$, $\pi(x, \varepsilon, u) \geq 0$.

6.5 Comparisons of risk aversion across agents

Definition 120. Let $\succsim^1$ and $\succsim^2$ be two risk averse EU maximizers. Then $\succsim^1$ is more risk averse than $\succsim^2$ if for each $F \in \Delta(Z)$ and each $c \in \mathbb{R}$,

$$F \succsim^1 \delta_c \text{ implies that } F \succsim^2 \delta_c.$$

Suppose that $\succsim^1$ and $\succsim^2$ are two risk averse EU maximizers and that $\succsim^1$ is more risk averse than $\succsim^2$. Let u_1 and u_2 be their respective utility indices. As we saw in the previous section, an agent is risk averse when his/her associated utility index is concave. As we will see, $\succsim^1$ is more risk averse than $\succsim^2$ whenever u_1 is “more concave” than u_2 .

The issue is then how to measure concavity of a function. We know that if the second derivative of a function is negative, then the function is concave. However, the comparison of second derivatives across functions does not tell us which function is more concave. Think for instance of u and $2u$. Clearly, these two functions represent the same agent. However, the absolute value of the second derivative of $2u$ is always twice that of u . This happens because the first derivative of $2u$ is always twice the first derivative of u . Thus, since the functions represent the same agent, the derivative of $2u$ has to decrease twice as fast as the derivative of u .

This is why in order to learn “how concave a function u is at x ,” one has to normalize the first derivative of u at x . Consider the function that assigns to each $t \in \mathbb{R}$ the value

$$\frac{1}{u'(x)}u(t).$$

This function’s derivative at $t = x$ is 1. Thus, its second derivative evaluated at $t = x$, i.e.,

$$\frac{1}{u'(x)}u''(x),$$

says how concave the function is at x , and can be compared to the second derivative of another function whose first derivative at x is also 1.

We want to define an index that tells us how concave the function is. Since the second derivative of concave function is non-positive, then we define the **coefficient of absolute risk aversion of u at x** as

$$r_A^u(x) \equiv -\frac{u''(x)}{u'(x)}.$$

Theorem 24. Let $\succsim^1$ and $\succsim^2$ be two risk averse EU maximizers with associated indices u_1 and u_2 , respectively. The following are equivalent.

1. $\succsim^1$ is more risk averse than $\succsim^2$.
2. For each $F \in \Delta(\mathbb{R})$ and each $c \in \mathbb{R}$,

$$\int u_1 dF \geq u_1(c) \text{ implies that } \int u_2 dF \geq u_2(c).$$

3. u_1 is more concave than u_2 , i.e., there is an increasing and concave function ψ such that $u_1 = \psi \circ u_2$.
4. If u_1 and u_2 are twice differentiable, then for each $x \in \mathbb{R}$,

$$r_A^{u_1}(x) \geq r_A^{u_2}(x).$$

5. For each $F \in \Delta(\mathbb{R})$, $c(F, u_1) \leq c(F, u_2)$.
6. For each $x \in \mathbb{R}$ and each $\varepsilon > 0$, $\pi(x, \varepsilon, u_1) \geq \pi(x, \varepsilon, u_2)$.

6.6 Comparisons of risk aversion across different levels of wealth

Let us consider a risk averse EU maximizer with associated utility index u . We are interested in studying when does the agent become less risk averse as he/she becomes wealthier. The answer depends on the nature of the risk the agent faces.

6.6.1 Decreasing risk aversion to additive risk

First we consider additive risks.

Let $w \in \mathbb{R}$ be the agent's initial wealth. Consider a random variable z with distribution F . We interpret the random variable as an additive lottery that will realize an additive gain for the agent. Then, the final distribution

of wealth of the agent is the random variable $w + z$ where z is distributed F . We denote the distribution of this random variable $w + F$.

The expected utility of $w + F$ is given by:

$$U(w + F) \equiv \int u(w + t) dF(t) = \int u(w + \cdot) dF.$$

That is, the expected utility of $w + F$ is also the expected utility of the random variable distributed F for the agent with utility function $u(w + \cdot)$, which is the horizontal translation of u to the left by w .

We are interested in characterizing the conditions under which the agent more easily accepts an additive lottery when his/her wealth increases. That is, when is it true that if the agent prefers the additive lottery F to the certain change in wealth c at a level of wealth w , then he/she will prefer F to the certain change in wealth c at any level of wealth higher than w .

Definition 121. Let $\succsim$ be a risk averse EU maximizer. Then $\succsim$ **exhibits decreasing risk aversion to additive risks** if for each $w_2 > w_1$, each $F \in \Delta(Z)$, and each $c \in \mathbb{R}$,

$$w_1 + F \succsim \delta_{w_1+c} \text{ implies that } w_2 + F \succsim \delta_{w_2+c}.$$

Theorem 25. The following are equivalent.

1. $\succsim$ exhibits decreasing risk aversion to additive risks.
2. For each $w_2 > w_1$, each $F \in \Delta(\mathbb{R})$, and each $c \in \mathbb{R}$,

$$\int u(w_1+t) dF(t) \geq u(w_1+c) \text{ implies that } \int u(w_2+t) dF(t) \geq u(w_2+c).$$

3. The EU preference associated with index $u(w_1 + \cdot)$ is more risk averse than the preference associated with index $u(w_2 + \cdot)$.
4. For each $w_2 > w_1$, the function $u(w_1 + \cdot)$ is more concave than $u(w_2 + \cdot)$. That is, there is an increasing and concave function ψ such that for each $t \in \mathbb{R}$, $u(w_1 + t) = \psi(u(w_2 + t))$.
5. If u is twice differentiable, the coefficient of absolute risk aversion for u is a non-increasing function, that is, if $x > y$, then

$$r_A^u(y) \geq r_A^u(x).$$

6. For each $w_2 > w_1$ and each $F \in \Delta(\mathbb{R})$,

$$c(F, u(w_1 + \cdot)) \leq c(F, u(w_2 + \cdot)).$$

Equivalently,

$$w_2 - c(w_2 + F, u) \leq w_1 - c(w_1 + F, u).$$

7. For each $w_2 > w_1$, each $t \in \mathbb{R}$, and each $\varepsilon > 0$,

$$\pi(t, \varepsilon, u(w_1 + \cdot)) \geq \pi(t, \varepsilon, u(w_2 + \cdot)).$$

Equivalently, for each $x > y$,

$$\pi(y, \varepsilon, u) \geq \pi(x, \varepsilon, u).$$

6.6.2 Decreasing risk aversion to multiplicative risk

Now we consider multiplicative risks. Throughout this subsection we assume that $Z \equiv \mathbb{R}_+$.

Let $w \in \mathbb{R}_+$ be the agent's initial wealth. Consider a random variable z with distribution F that has support on $[0, +\infty)$. We interpret the random variable as a multiplicative lottery that will realize a multiplicative gain for the agent. For instance if z turns out to be 2, then the agent's wealth is multiplied by 2. Then the final distribution of wealth of the agent is the random variable wz where z is distributed F . We denote the distribution of this random variable wF .

The expected utility of wF is given by:

$$U(wF) \equiv \int u(wt) dF(t) = \int u(w \cdot) dF.$$

That is, the expected utility of wF is also the expected utility of the random variable distributed F for the agent with utility function $u(w \cdot)$, which is the horizontal contraction of u to the left by the ratio $\frac{1}{w}$.

We are interested in characterizing the conditions under which the agent more easily accepts a multiplicative lottery when his/her wealth increases. That is, when is it true that if the agent prefers the multiplicative lottery F to the certain multiplicative change in wealth c at a level of wealth w , then he/she will prefer F to the certain multiplicative change in wealth c at any level of wealth higher than w .

Definition 122. Let $\succsim$ be a risk averse EU maximizer. Then $\succsim$ **exhibits decreasing risk aversion to multiplicative risks** if for each $w_2 > w_1 \geq 0$, each $F \in \Delta(\mathbb{R})$ with support $[0, +\infty)$, and each $c \in \mathbb{R}_+$,

$$w_1 F \succsim \delta_{w_1 c} \text{ implies that } w_2 F \succsim \delta_{w_2 c}.$$

Theorem 26. The following are equivalent.

1. $\succsim$ exhibits decreasing risk aversion to multiplicative risks.
2. For each $w_2 > w_1$, each F , and each $c \in \mathbb{R}$,

$$\int u(w_1 t) dF(t) \geq u(w_1 c) \text{ implies that } \int u(w_2 t) dF(t) \geq u(w_2 c).$$

3. For each $w_2 > w_1 \geq 0$, The EU preference associated with index $u(w_1 \cdot)$ is more risk averse than the preference associated with index $u(w_2 \cdot)$.
4. For each $w_2 > w_1 \geq 0$, the function $u(w_1 \cdot)$ is more concave than $u(w_2 \cdot)$. That is, there is an increasing and concave function ψ such that for each $t \in \mathbb{R}$, $u(w_1 t) = \psi(u(w_2 t))$.
5. If u is twice differentiable, then for each $t \in \mathbb{R}_+$,

$$r_A^{u(w_1 \cdot)}(t) \geq r_A^{u(w_2 \cdot)}(t).$$

6. If u is twice differentiable, then the coefficient of relative risk aversion for u defined by

$$r_R^u(x) \equiv -x \frac{u''(x)}{u'(x)},$$

is a non-increasing function.

7. For each $w_2 > w_1 \geq 0$ and each $F \in \Delta(\mathbb{R})$ with support $[0, +\infty)$,

$$c(F, u(w_1 \cdot)) \leq c(F, u(w_2 \cdot)).$$

Equivalently,

$$\frac{c(w_1 F, u)}{w_1} \leq \frac{c(w_2 F, u)}{w_2}.$$

6.7 Problems

1. Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be a non-decreasing and concave function. Prove that for each $x \in \mathbb{R}$, $r_A(x, u) = a > 0$, if and only if u is an affine linear transformation of the function $x \in \mathbb{R} \mapsto \tilde{u}(x) \equiv -e^{-ax}$.
2. Suppose that there is a risk free asset that returns $r > 0$ dollars for each dollar invested and a risky asset that for each dollar invested returns 0 dollars with probability $\frac{1}{2}$ and $3r > 0$ dollars with probability $\frac{1}{2}$. Let $\succsim_1$ and $\succsim_2$ be two expected utility maximizers with indices u_1 and u_2 respectively. Assume that both agents are risk averse and that each of them has a budget of one hundred dollars to allocate between these two assets.
 - (a) Characterize the agents' optimal portfolio.
 - (b) Prove that at their optimal portfolio, the agents buy a non-zero amount of the risky asset.
 - (c) Prove that if agent 1 is more risk averse than agent 2, then at her optimal portfolio, agent 2's optimal share of the risky asset is greater than or equal to agent 1's.
3. An Expected Utility preference on $\Delta(Z)$ is **risk neutral** if for each $F \in \Delta(Z)$, $F \sim \delta_{\mu_F}$. Prove that $\succsim$ is risk neutral if and only if it is represented by the EU associated with each index u of the form $u(t) = at + b$ where $\{a, b\} \subseteq \mathbb{R}$ and $a > 0$. Conclude that there is only one risk neutral preference and it is represented by the EU whose associated index is $t \in \mathbb{R} \mapsto t$. Thus, the risk neutral preference $\succsim$ is the one that for each $\{F, G\} \subseteq \Delta(Z)$, $F \succsim G$ if and only if $\mu_F \geq \mu_G$.
4. An asset is a divisible claim to a financial return in the future. Suppose that there are two assets, a safe asset with a return of 1 dollar per dollar invested and a risky asset with a random return of z dollars per dollar invested. The random return z has the following probability distribution: $z = 0$ with probability α , $z = 1$ with probability β , and $z = 2$ with probability $1 - \alpha - \beta$.

A **risk neutral** individual, who we refer to as agent A , has an initial wealth w to invest in these two assets. Let x be the part of agent A 's wealth invested in the risky asset. The amount invested in the safe asset is $w - x$. Thus, for any realization the random return z , agent A 's portfolio $(x, w - x)$ pays $xz + w - x$.

- (a) Characterize agent A 's optimal portfolio, i.e., find x in terms of α and β . When does agent A invest in the risky asset? Interpret.
 - (b) Suppose now that agent A cannot directly invest in these two assets. Instead, agent A can invest through a financial manager who gets a percentage of the agent's return. Suppose for simplicity that agent A would receive a return of zero if she does not invest through a financial manager. There are two financial managers: $M1$ and $M2$. Suppose that the initial wealth of $M1$ and $M2$ is zero and that each gets a portion $\varphi \in (0, 1)$ of the corresponding agents' returns. For instance, if agent A gives y_1 dollars to $M1$ and this financial manager invests x_1 in the safe asset and $y_1 - x_1$ in the risky asset, then for each realization of the random return, the manager gets $\varphi(x_1 z + y_1 - x_1)$ and agent A gets $(1 - \varphi)(x_1 z + y_1 - x_1)$.
 - i. Suppose that $M1$ is an expected utility maximizer with vNM utility index: $u_1(x) \equiv \sqrt{x}$. If agent A gives y_1 dollars to $M1$, what is the managers' optimal portfolio? That is, calculate x_1 in terms of α and β . When does $M1$ invest in the risky asset?
 - ii. Suppose that $M2$ is an expected utility maximizer with vNM utility index: $u_2(x) \equiv \ln x$. If agent A gives y_2 dollars to $M2$, what is the managers' optimal portfolio? That is, calculate x_2 in terms of α and β . When does $M2$ invest in the risky asset?
 - iii. Characterize agent A 's optimal portfolio, i.e., find the amounts y_1 and y_2 that agent A gives to $M1$ and $M2$, respectively. In a graph of α vs. β show the different portfolios associated with the pairs (α, β) .
5. Suppose that there is a risk free asset that returns $r > 0$ dollars for each dollar invested and a risky asset that for each dollar invested returns 0 dollars with probability $\frac{1}{2}$ and $3r > 0$ dollars with probability $\frac{1}{2}$. Let $\succsim$ be an expected utility maximizer whose associated index is u . Assume that the agent is risk averse and has x dollars to allocate between these two assets.
- (a) Characterize the agents' optimal portfolio.
 - (b) Assume that the agent exhibits decreasing relative risk aversion. Prove that her demand of the risky asset as a proportion of total wealth increases with wealth.

- (c) Select u that exhibits constant relative risk aversion (find one). Calculate the demand of the risky asset as a proportion of total wealth. Is it constant?
- 6. An expected utility maximizer over lotteries of money has utility index u and initial wealth w . Lottery l offers a payoff of $x > 0$ dollars with probability α and a payoff of $y > x$ dollars with probability $1 - \alpha$.
 - (a) If the agent owns the lottery, what is the minimum price $\underline{p}$ at which she is willing to sell it (find the equation that characterizes $\bar{p}$).
 - (b) If the agent does not own the lottery, what is the maximum price $\underline{p}$ that she is willing to pay for it (find the equation that characterizes $\underline{p}$).
 - (c) If the agent is risk averse and exhibits non-increasing absolute risk aversion, what can we say about the relation between $\underline{p}$ and $\bar{p}$?
 - (d) If the agent is risk averse and exhibits non-increasing relative risk aversion, what can we say about the relation between $\underline{p}$ and $\bar{p}$?

6.8 Comparison of distributions in terms of return and risk

When is a lottery F unambiguously better than a lottery G for a money monotone EU maximizer? For example if $x > y$, we know that for each EU money monotone preference $\succsim$,

$$\delta_x \succsim \delta_y.$$

We now characterize these situations.

Definition 123. Let $\{F, G\} \subseteq \Delta(\mathbb{R})$. Then, F **first order stochastically dominates** G , denoted $F \geq_{FSD} G$ if for each money monotone EU preference $\succsim$,

$$F \succsim G.$$

Equivalently, if for each non-decreasing $u : \mathbb{R} \rightarrow \mathbb{R}$,

$$E_F u \geq E_G u.$$

Definition 124. A lottery $F \in \Delta(\mathbb{R})$ has compact support if there is $m \in \mathbb{R}$ such that $F(-m) = 0$ and $F(m) = 1$.

Proposition 61. Let $\{F, G\} \subseteq \Delta(\mathbb{R})$ with compact support. Then, $F \geq_{FSD} G$ if and only if for each $x \in \mathbb{R}$, $F(x) \leq G(x)$.

Proof. ($\Rightarrow$) Suppose that $F \geq_{FSD} G$. Let $x \in \mathbb{R}$. We prove that $F(x) \leq G(x)$. Consider the step function

$$u(t) \equiv \begin{cases} 0 & \text{if } t \leq x \\ 1 & \text{otherwise.} \end{cases}$$

Since u is non-decreasing, then $E_F u \leq E_G u$. Then,

$$\int u dF \geq \int u dG.$$

Equivalently,

$$1 - F(x) \geq 1 - G(x).$$

That is, $F(x) \leq G(x)$.

($\Leftarrow$) Suppose that for each $x \in \mathbb{R}$, $F(x) \leq G(x)$. Let u be a non-decreasing function. We claim that $E_F u \geq E_G u$.

For each $x \in \mathbb{R}$, let $H(x) \equiv F(x) - G(x)$. Then, integrating by parts, for m large enough

$$\int_{-\infty}^{\infty} u dH = u(x)H(x)|_{-m}^m - \int_{-\infty}^{\infty} H(t)u'(t)dt \geq 0,$$

since $u' \geq 0$ and $H(-m) = H(m) = 0$. Thus,

$$\int u dF \geq \int u dG.$$

□

Now we consider the question: when is a lottery F unambiguously riskier than another lottery G ?

Definition 125. Let $\{F, G\} \subseteq \Delta(\mathbb{R})$. Then, F **second order stochastically dominates** G , denoted $F \geq_{SSD} G$ if for each money monotone risk averse EU preference $\succsim$,

$$F \succsim G.$$

Equivalently, if for each concave and non-decreasing $u : \mathbb{R} \rightarrow \mathbb{R}$,

$$E_F u \geq E_G u.$$

Proposition 62. Let $\{F, G\} \subseteq \Delta(\mathbb{R})$ with compact support be such that $\mu_F = \mu_G$. Then, $F \geq_{SSD} G$ if and only if for each $x \in \mathbb{R}$,

$$\int_{-\infty}^x F(t) dt \leq \int_{-\infty}^x G(t) dt.$$

Definition 126. Let $\{F, G\} \subseteq \Delta(\mathbb{R})$. Then, G is a **mean preserving spread** of F if there is a family of distributions $\{H_x : x \in \mathbb{R}, \mu_{H_x} = 0\}$ such that for each $y \in \mathbb{R}$,

$$G(y) = \int_{-\infty}^y \left(\int_{-\infty}^{y-x} dH_x(t) \right) dF(x).$$

Proposition 63. Let $\{F, G\} \subseteq \Delta(\mathbb{R})$ with compact support be such that $\mu_F = \mu_G$. Then, $F \geq_{SSD} G$ if and only if G is a mean preserving spread of F .

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